

## Project 3

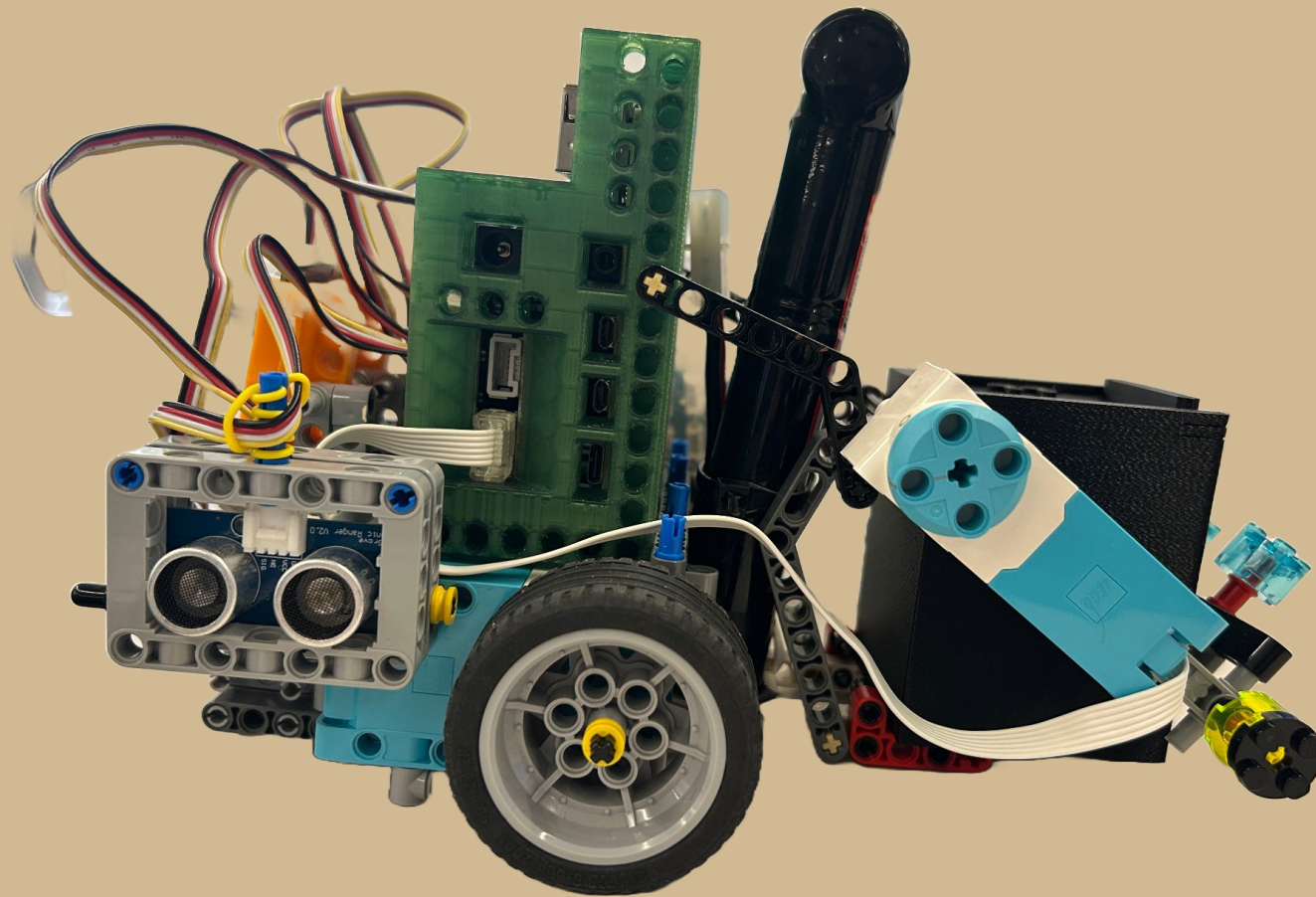
# Global Emergency Autonomous Response System Recommendation (Technical Sections)

## Team 11

Hanna Bowman, Luca DalCanto, Owen Goudelock, Lilja Kiiski



# Final Design



# **Project Goals & Management**

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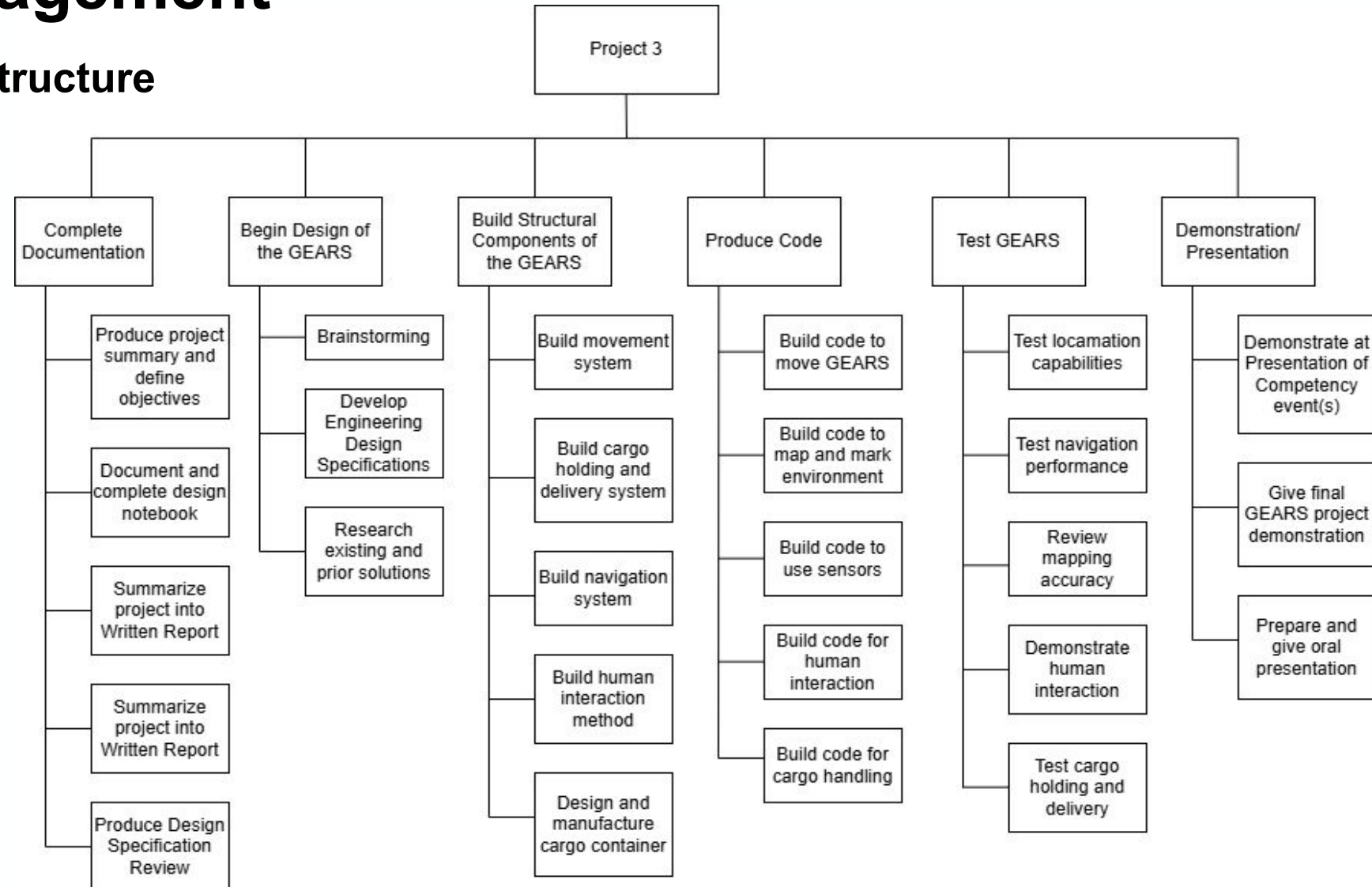
# Project Goals

## Technical Requirements

| Customer Need                                         | Technical Need                                                                                                                     | Technical Requirement | Target Value |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|
| Precise area navigation                               | Do not come within certain distance of all hazards                                                                                 | $\geq 2\text{cm}$     | 4cm          |
| Timely cargo transport                                | Total mission time (transport and return)                                                                                          | $\leq 18$ minutes     | 12 minutes   |
| Detect and characterize hazardous location in a table | Detect a certain percentage of all hazards and display in table                                                                    | $\geq 90\%$           | 100%         |
| Generate a map of junctions in area                   | Generate a map of a certain percentage of accuracy displaying junctions                                                            | $\geq 90\%$           | 100%         |
| Product is culturally considerate                     | Design uses language independent communication in symbols and human-interaction components in a certain percentage of its displays | $\geq 95\%$           | 100%         |
| Cargo delivered status indication                     | Indicate the cargo delivered status in non-hostile manner so that a person from a certain distance away can notice indication      | $\geq 15\text{in}$    | 20in         |

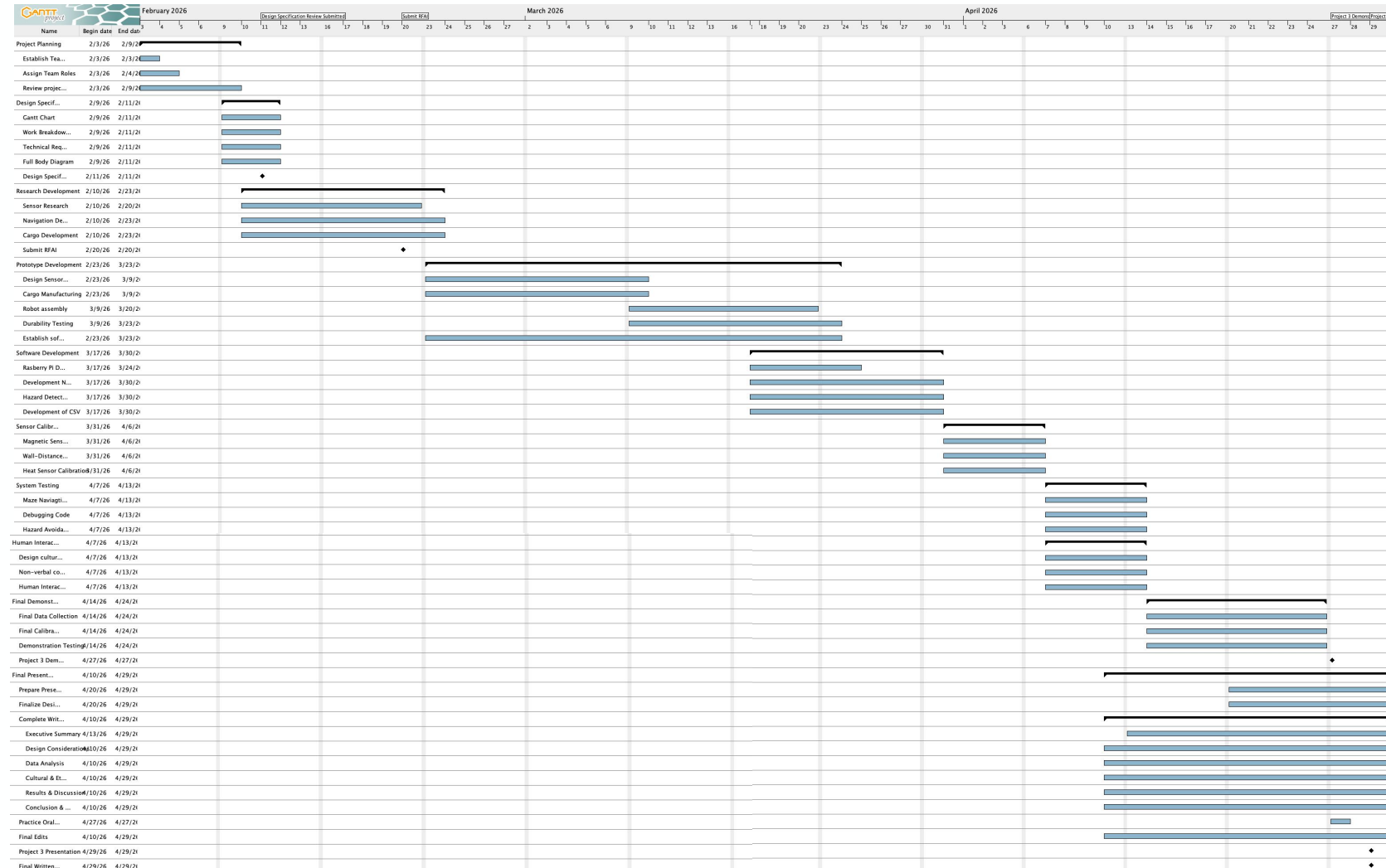
# Project Management

## Work Breakdown Structure



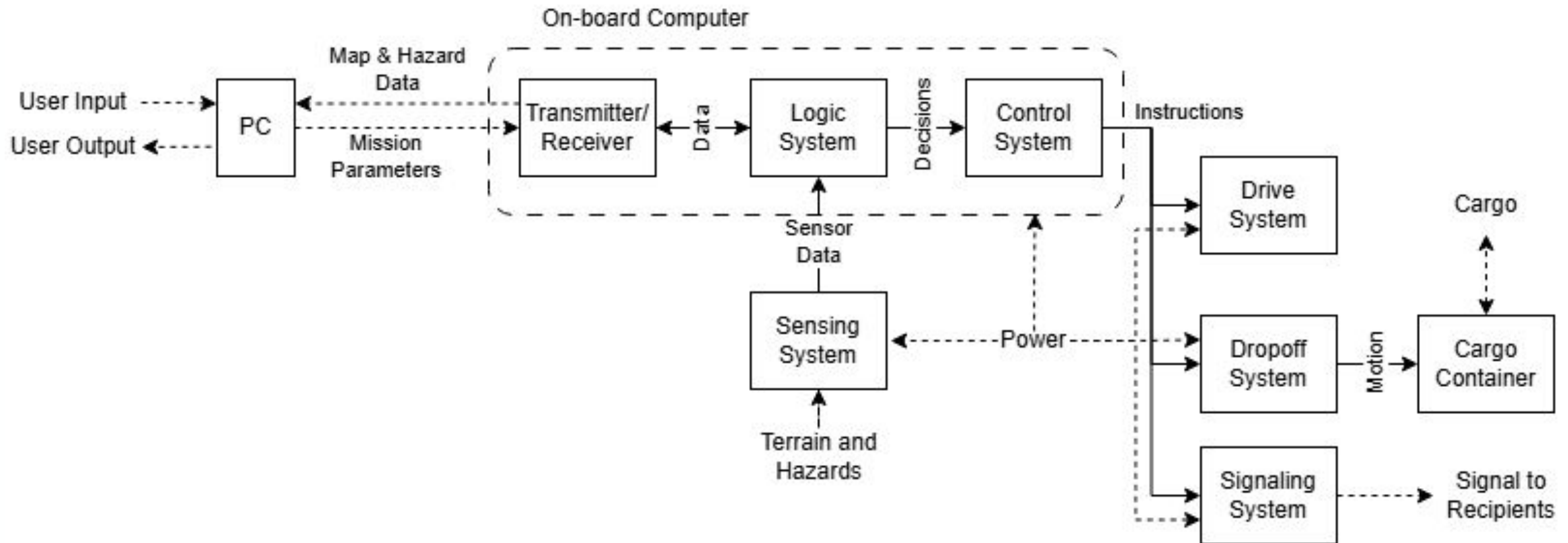
# Project Management

## Gantt Chart



# Design Overview

## Functional Block Diagram



# **Robot Design Philosophy: Mechanics**

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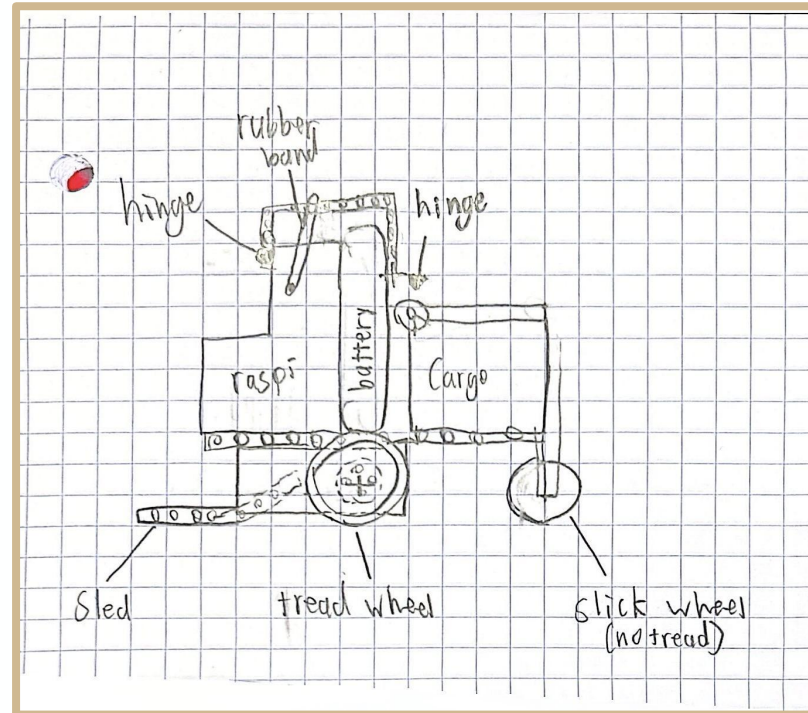
# Prototyping

## Hardware

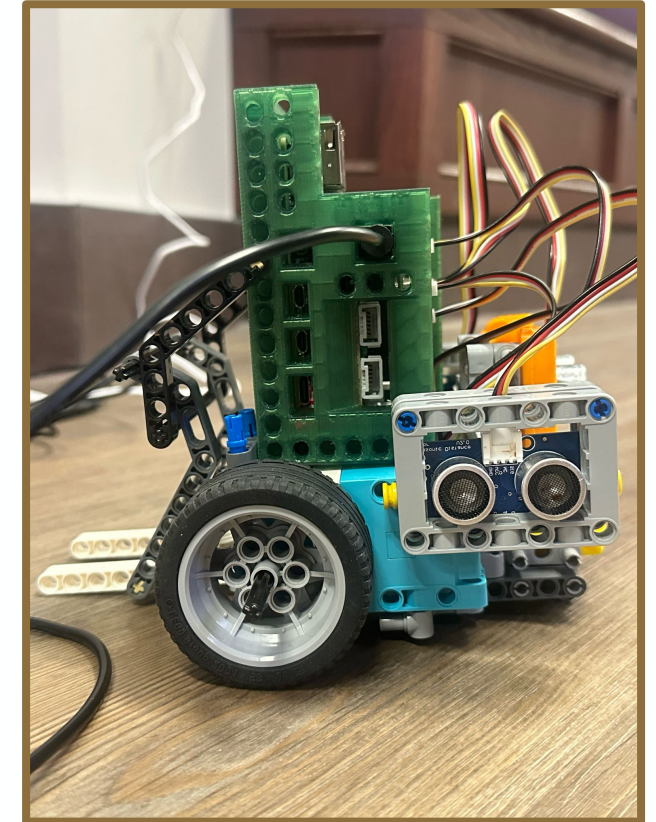
### Needs:

- Maneuverability
- Compactness
- Ample sensors
- Cargo dropoff

### Conceptual Sketch



### First Prototype



# Sensor Base

## Ultrasonic (3):

- Distance measurement
- Front is mounted higher

## Infrared:

- Measures two analogue voltage values
- Identifies dangerous heat sources

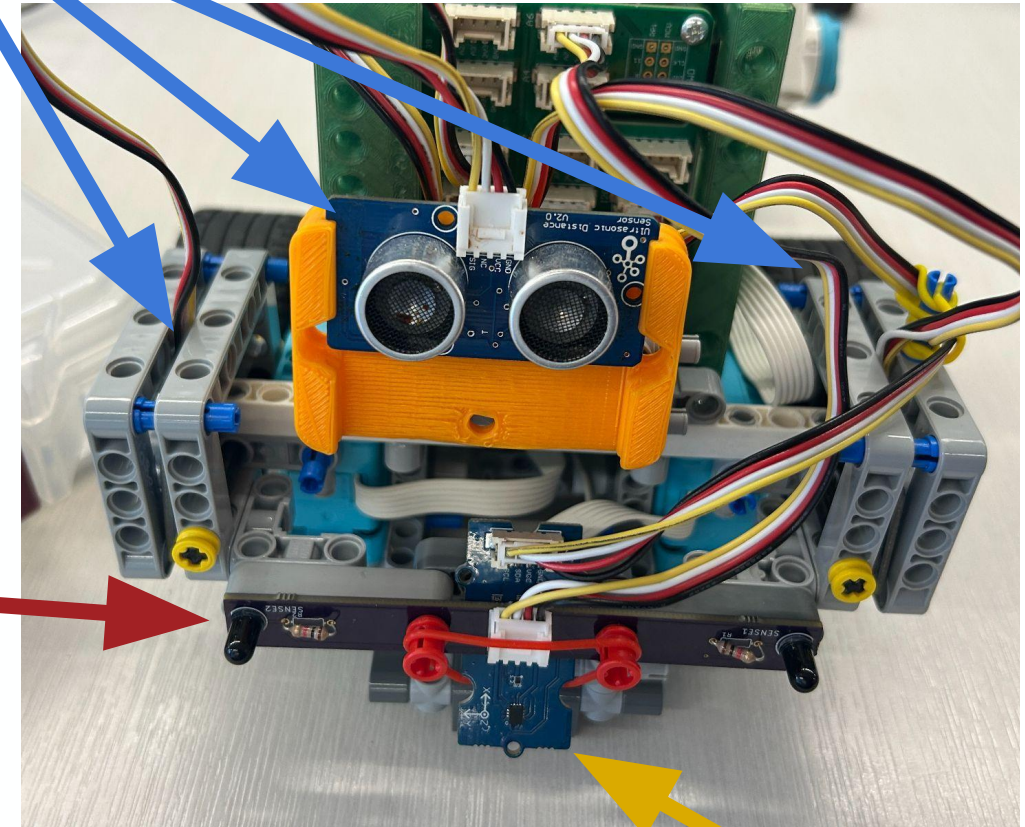
## IMU:

- Measures inertial movement and magnetic fields
- Inertial was deemed too faulty
- Magnetic sense was used

Ultrasonic

Infrared

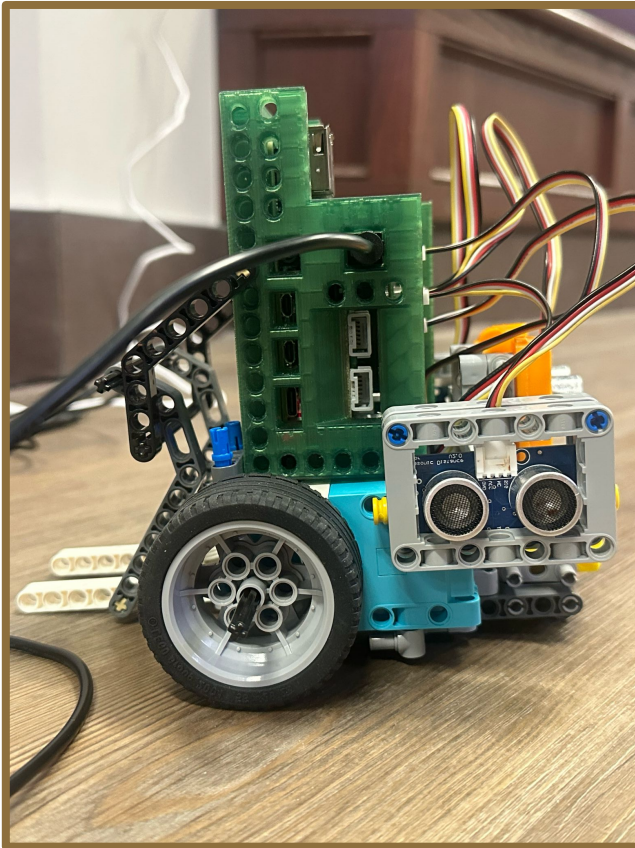
IMU



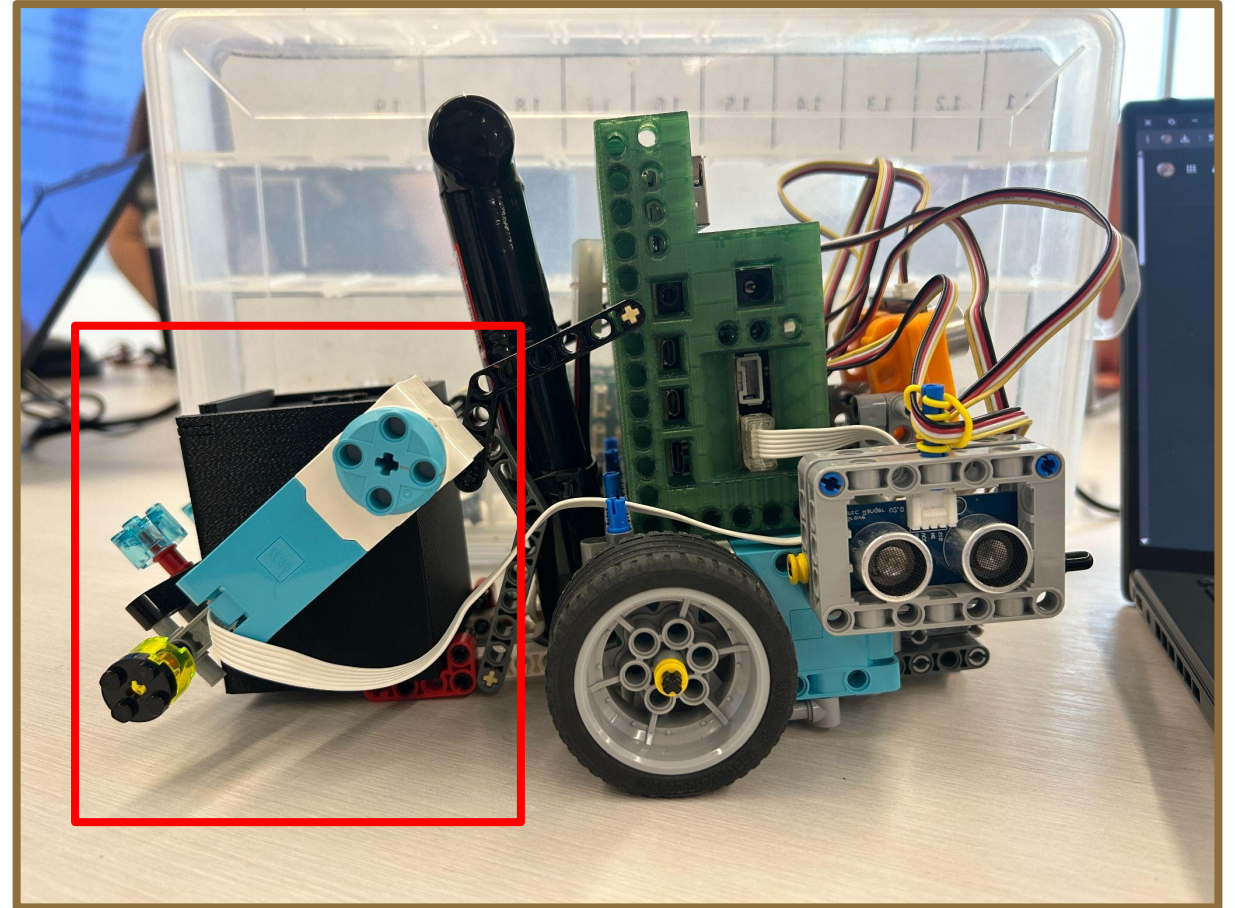


# Design Changes

POC #1 Prototype



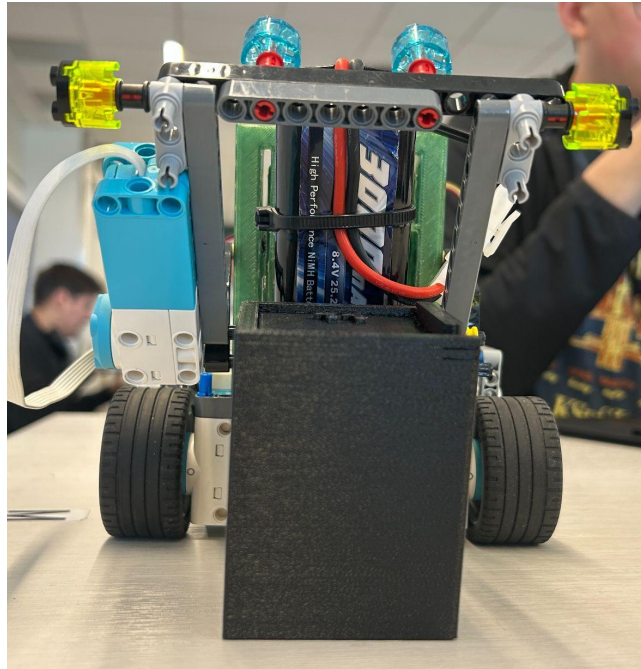
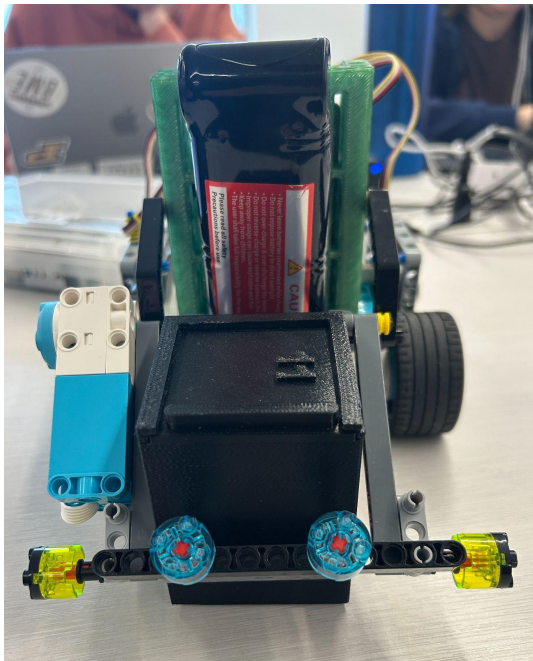
Final Demo GEARE



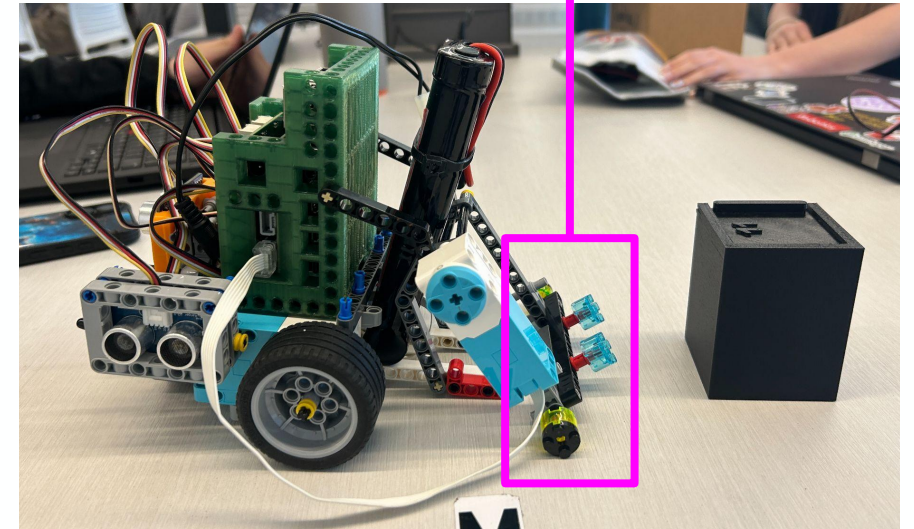


# Cargo

## Drop Off Mechanism

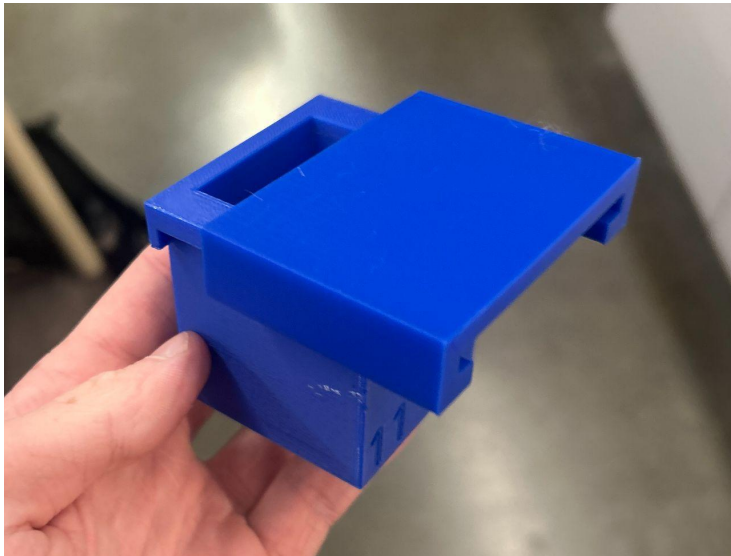


Illumination



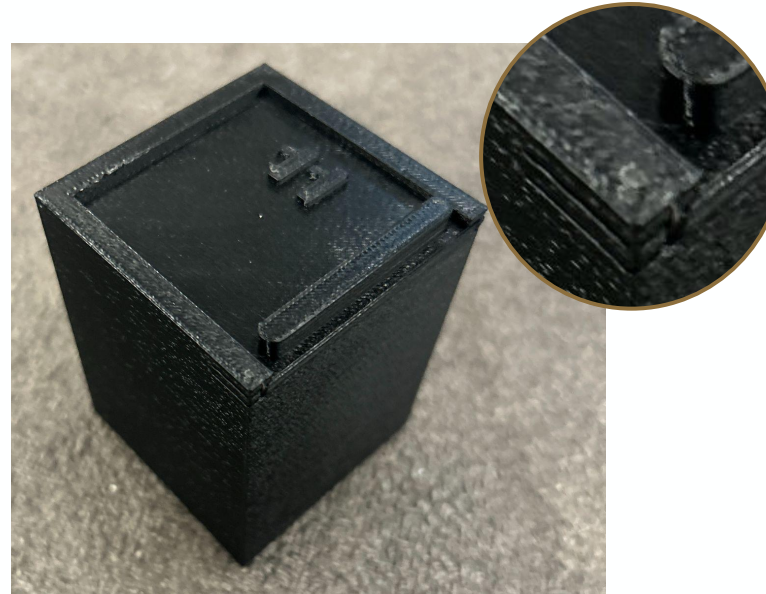
# Cargo

## Comparing three iterations



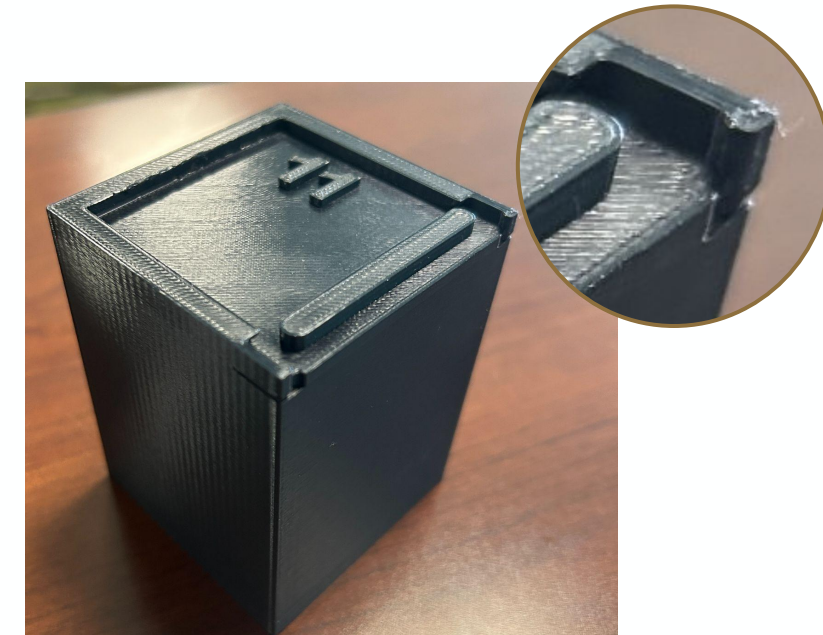
**First (OnShape) Blue PLA+**

- Tight tolerance
- Small capacity
- Irregular shape
- Unclear opening



**Second (NX) Black PLA+**

- Good tolerance
- Clear handle
- Fragile locking mechanism
- Loosened over time



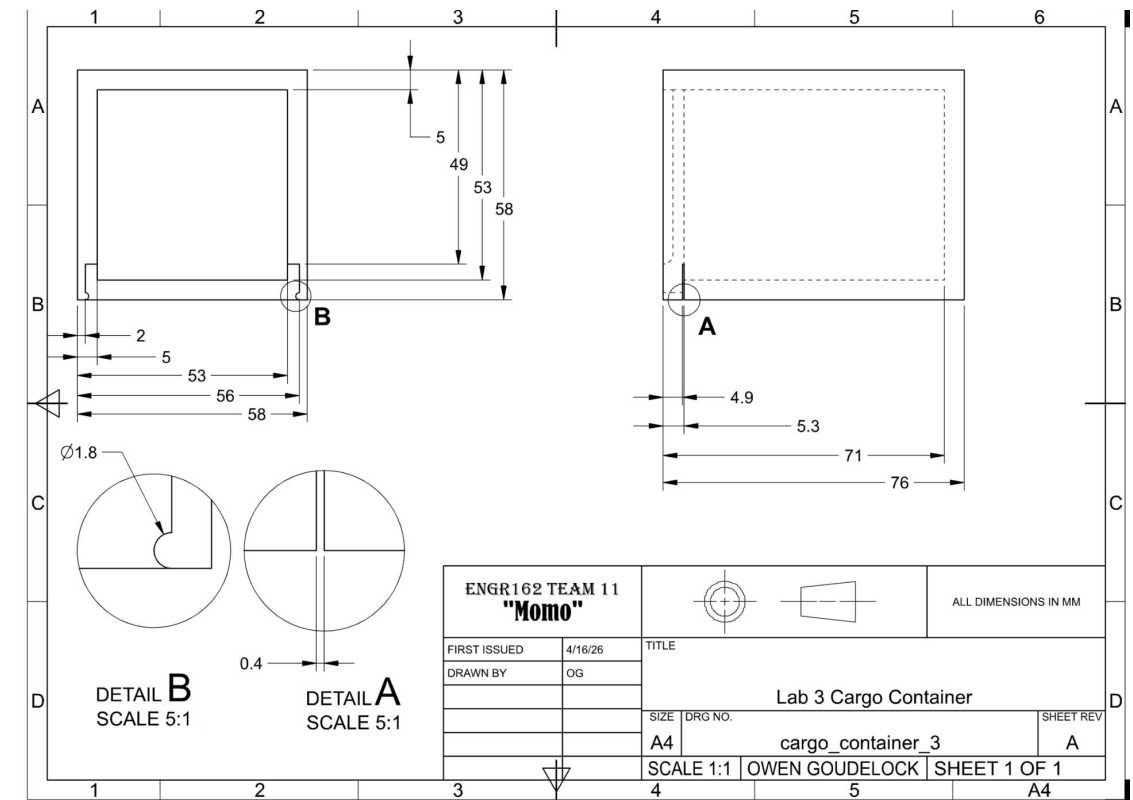
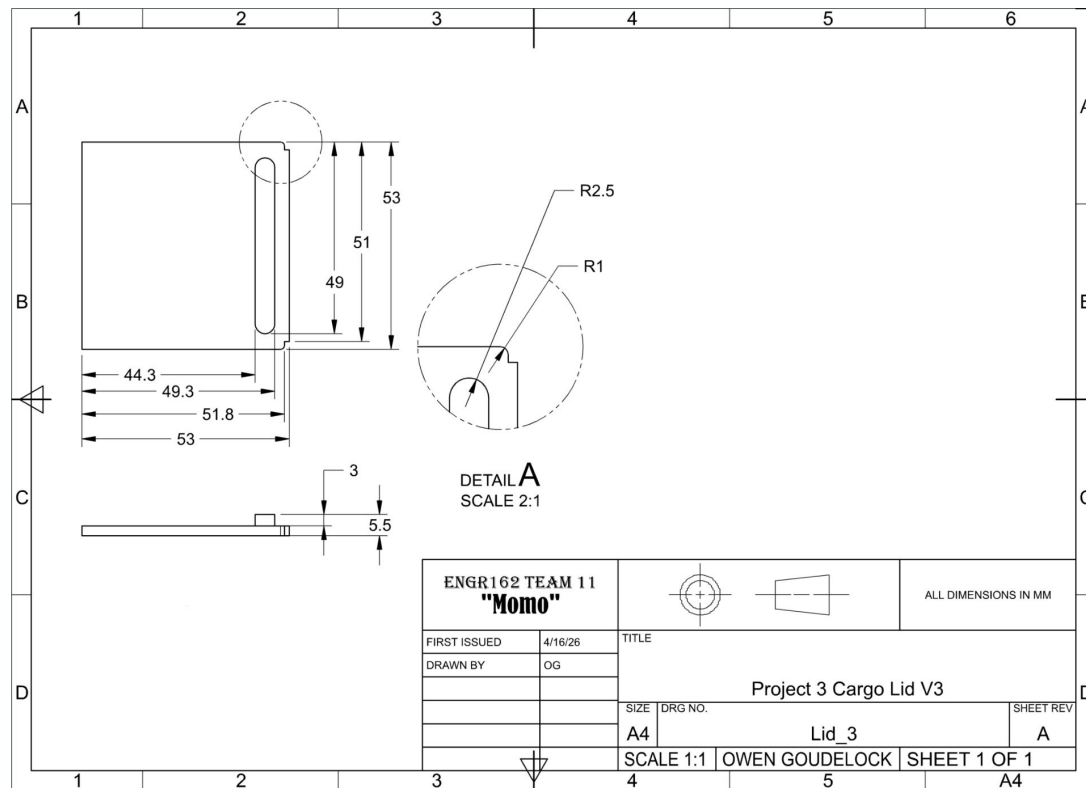
**Final (NX) Black PLA+**

- Good tolerance
- Flush handle
- Strong locking mechanism

# Cargo

## Final design

0.3 mm vertical clearance, intentional 0.8 mm overlap for compliant lock

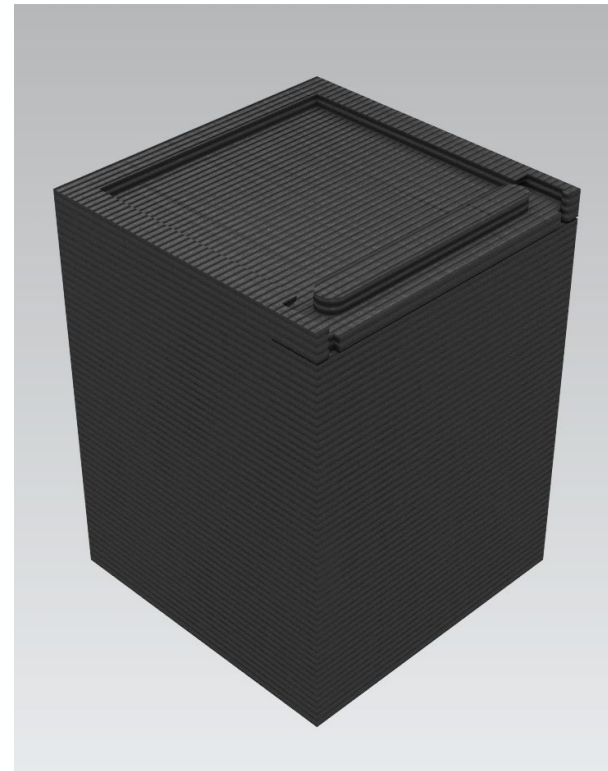
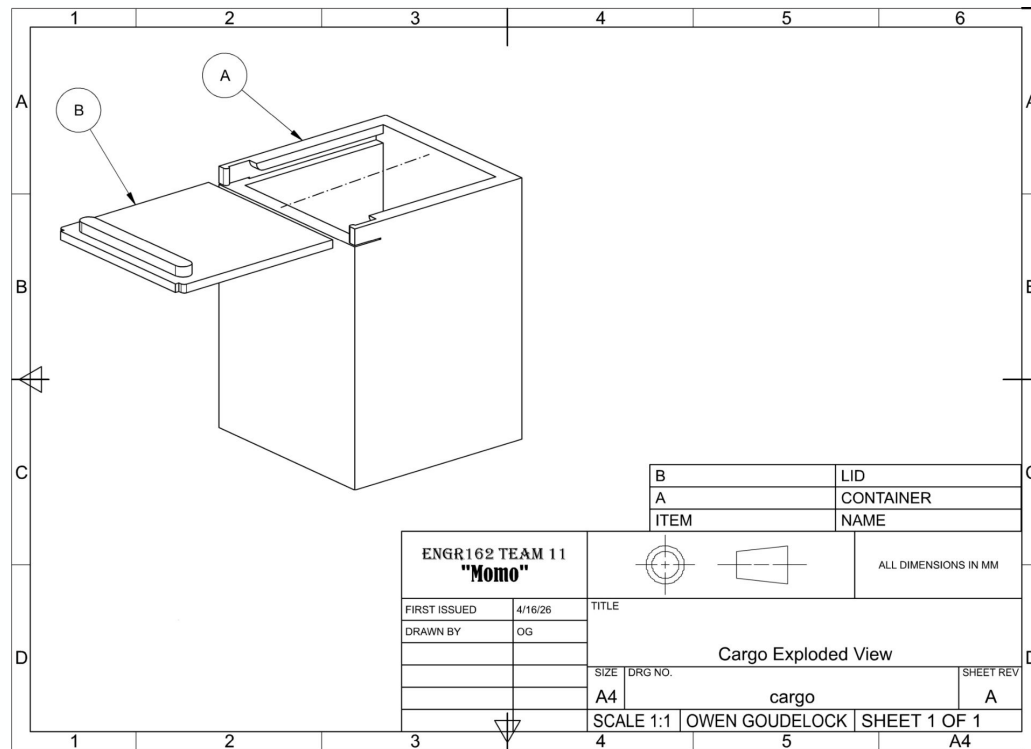




# Cargo

## Final design

Black PLA+, Prusa Mk4s @ 0.2 mm layer height



# **Robot Design Philosophy: Software**

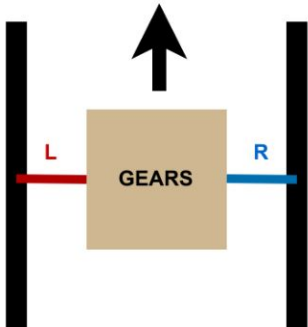
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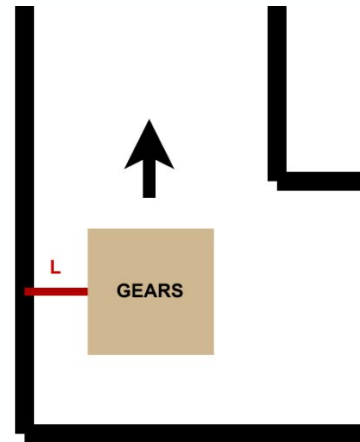
# Driving: Robot Movement & PID

## Wall Following

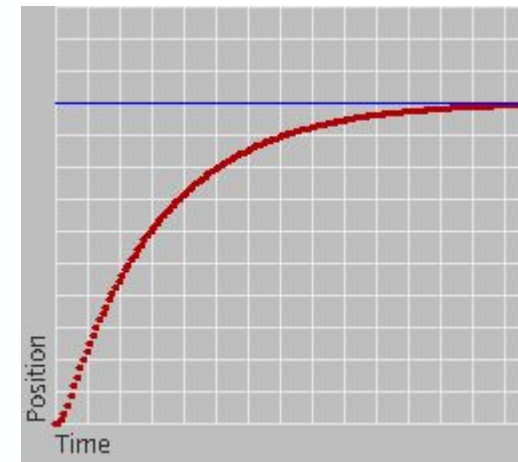
- Robot takes initial measurements from left & right ultrasonics to calculate track width (excluding robot width)
- Robot then uses a PD loop (P: 0.1 and D: 0.005) to veer left/right in order to stay centered



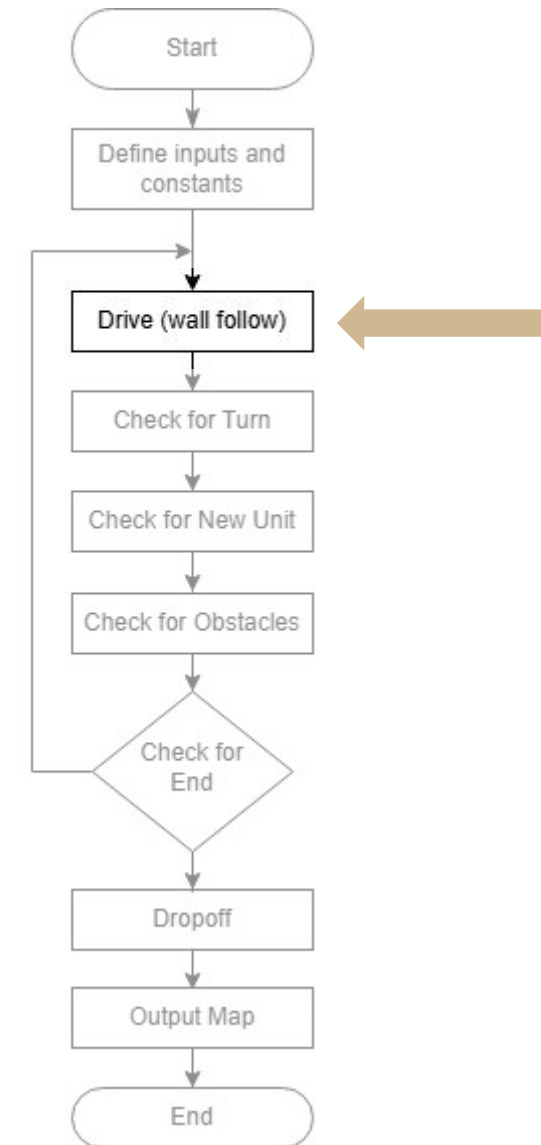
With both walls present -- error from **left AND right** distance



With one wall present -- error from track width and **left OR right** distance



Approximate best-case scenario for PID action



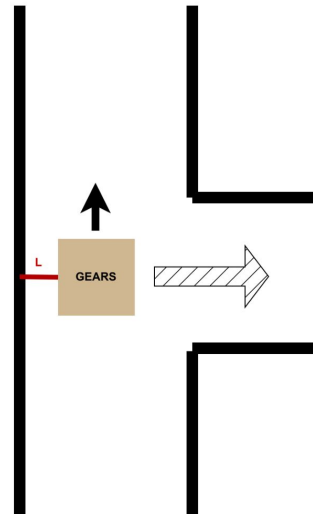
# Maze Solving

## Robot Turning

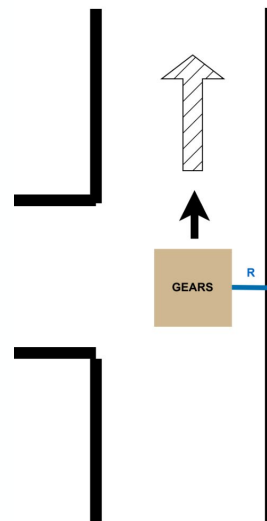
To ensure the maze is solved, the GEARS takes actions with this priority scheme:

1. Right turn, if possible
2. Straightaway, if no right turn
3. Left turn, if only option

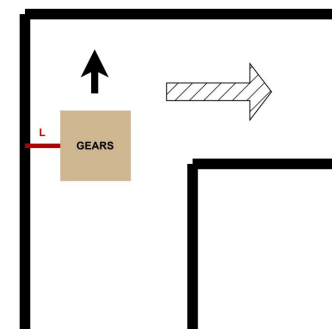
Types of turns:



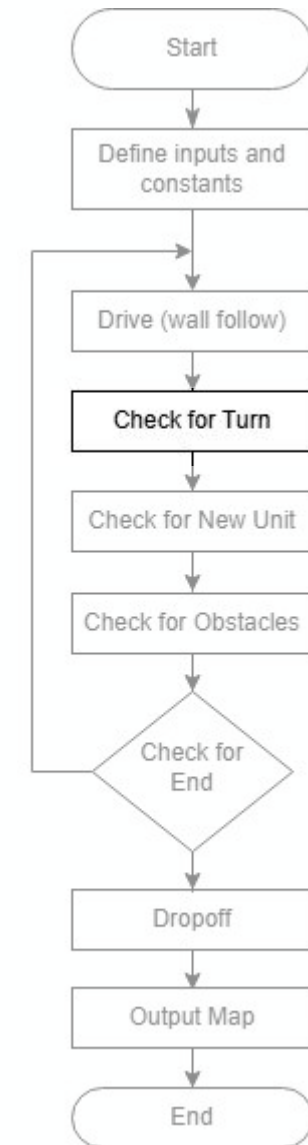
Right-Straight



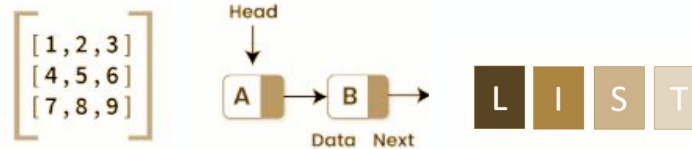
Left-Straight



Right/Left/Right-Left



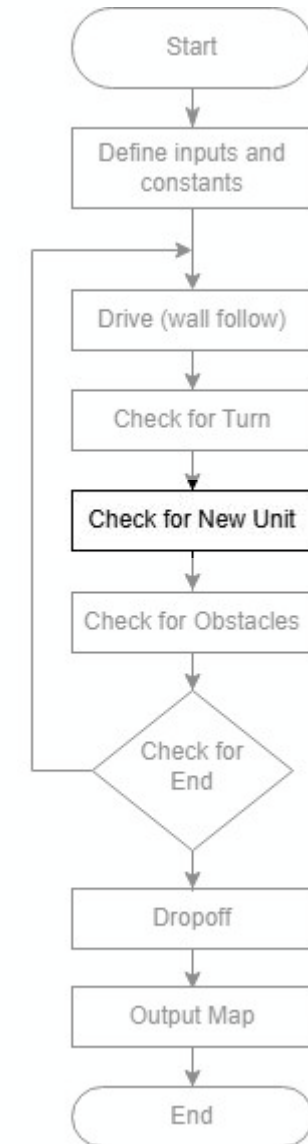
# Mapping Implementation & Decision Matrix



| Criteria                        | Weight | 2D Array (Grid) | Singly Linked List | 1D List of Coordinates |
|---------------------------------|--------|-----------------|--------------------|------------------------|
| Simplicity of Implementation    | 0.4    | 2.4             | 0.8                | 3.6                    |
| Allows for Relative Coordinates | 0.3    | 0.9             | 2.4                | 2.1                    |
| Ease of Debugging               | 0.3    | 1.8             | 0.9                | 2.4                    |
| Total                           | 1      | 5.1             | 4.1                | 8.1                    |

```
class GridSquare:
    """
    x = coordinate x
    y = coordinate y
    type = one of the following options
    > 0 : unknown
    > 1 : traversed square
    > 2 : heat source
    > 3 : mag source
    > 4 : end
    > 5 : origin
    """
    def __init__(self, x, y, typo, value=-1):
        self.x = x
        self.y = y
        self.typo = typo
        self.value = value
```

GridSquare class implemented to store information for each grid space detected / travelled to

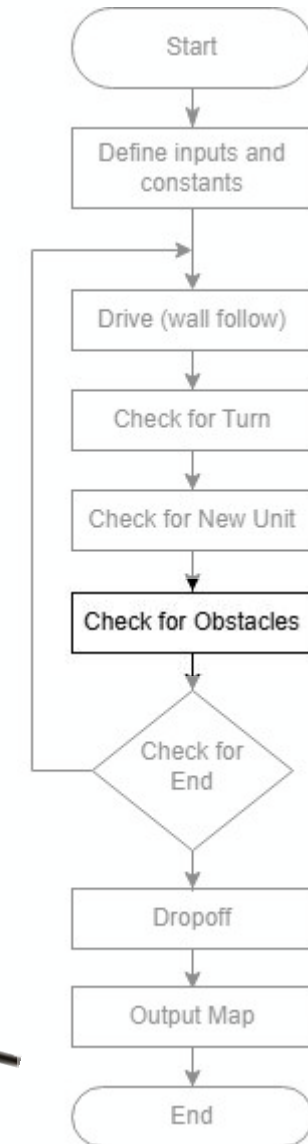


# Magnet and Infrared Sensing

## Detection and Robot Actions

- Primary hazards faced by GEARS:
  - High temperature heat sources from fires
  - High electrical and magnetic activity from damaged power stations.
- Inertial Measurement Unit (IMU) and Infrared (IR) sensors used
- Code is run every cycle to detect heat/magnetic sources
  - Magnet & IR each have specific threshold values
  - If value is passed, code registers as “detected”
- If detected, source value is recorded and placed on map
  - Robot immediately backtracks to last unit center
  - Then forces turn (90, -90, 180)

(Images are not to scale)





# Dropoff & Display

## End of the Run

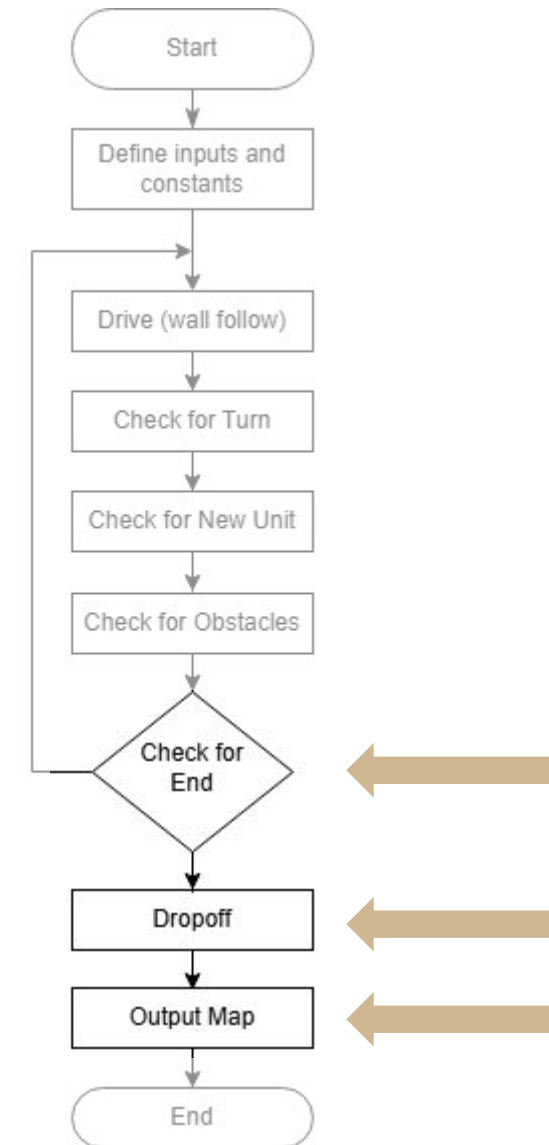
- **Exit condition: all ultrasonic sensors detect NONE or large distance**
- Once exit is detected, cargo is released, and robot retreats away from cargo to indicate completion.
- The list of GridSquares must be converted to two CSV files:

### Map CSV

1. Define 2D array size based on max and min coordinate values
2. For each GridSquare, populate its value in the 2D array
3. Fill the rest with zeros
4. Invert the rows of the 2D array
5. Add headers and export to CSV

### Hazard CSV

1. Write headers to CSV
2. For each hazard GridSquare, write its details to the CSV



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# Software Design Process and Iteration

## Software

- Team implemented various components of robot software in the following approximate order
  - Basic driving
  - PID-based driving correction
  - Maze Solving & Turning
  - Cargo Dropoff & Mapping
- Each component was tested on its own before integration with other subcomponents
- Team tested iteratively for thresholds for sensor readings, and constants for PID
  - ie. moving robot near sensor to read values
- Team had challenges with various steps of the process
  - Fine-tuning angle for turning
    - Required adding an extra “push” depending on 90/180/270/360 turn
  - Faulty sensor readings
    - Came up with strategies to combat poor values given from IR and IMU sensors

# Conclusions

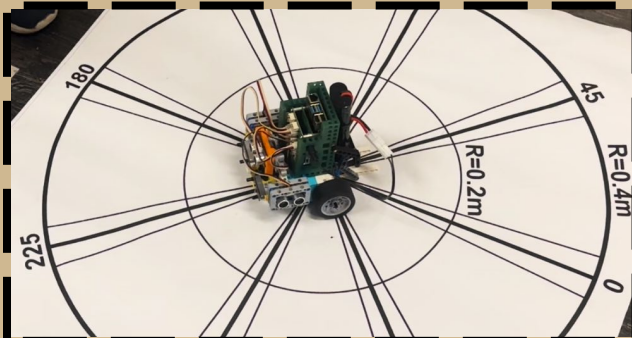
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# Demonstration Performance

## Quantitative Evidence for Success

### POC Individual Tasks

**6/6**



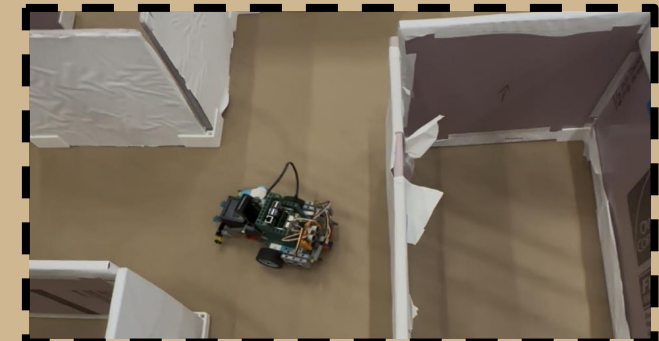
### POC Integrated Tasks

**3/4**



### Final Demo

**Failed to  
complete 2/3  
maze  
segments**



# Positive Attributes

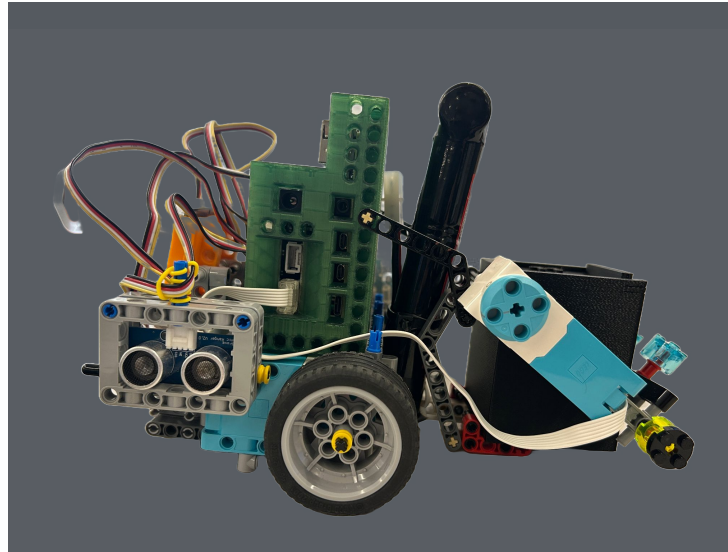
## Attributes to be proud of with GEARS Robot

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 0 | 0 | 3 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 2 | 1 | 0 | 1 | 1 | 0 |
| 0 | 4 | 0 | 5 | 0 | 0 |

Figure 2: Example map

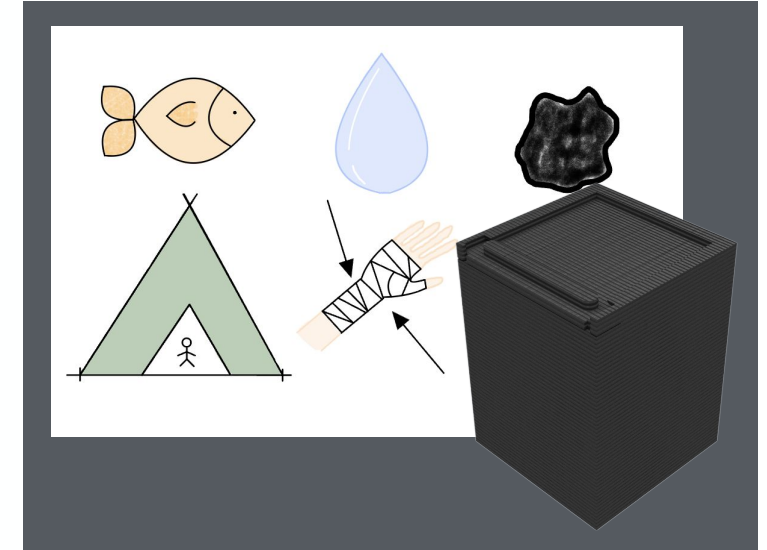
### Map Generation

Robot had consistent success with creating an accurate map of the locations



### Structural Simplicity

Final GEARS Robot is very compact, light, and thus easy to transport



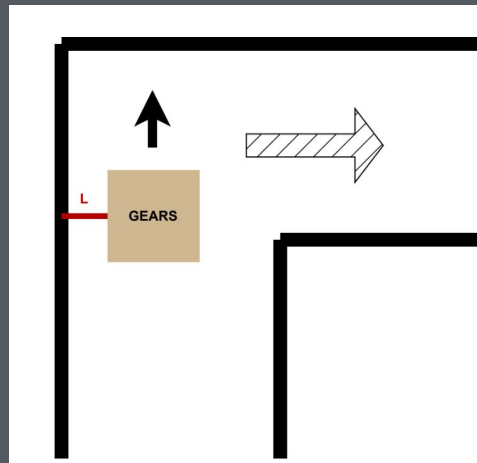
### Cultural Symbols & Cargo

Team put in effort to make a sturdy cargo container and culturally adequate symbols



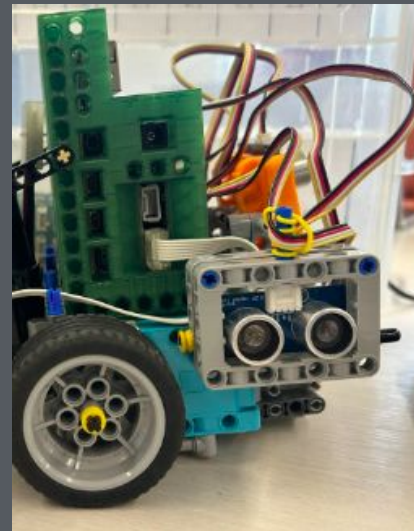
# Negative Attributes

## Attributes to be improved with GEARS Robot



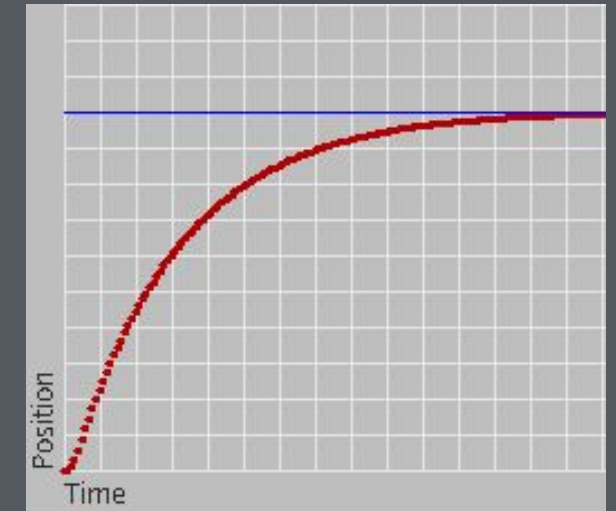
### Error Accumulation

Final GEARS Robot was +/- 10-20° off with turning on some occasions



### Ultrasonic Placements

GEARS Robot on occasion detected magnet / IR sensor as a wall before detecting its sensor values, due to ultrasonics being closer to the ground



### PID Fine-Tuning

While the PID used was mostly sufficient, should be fine tuned to prevent crashing into walls

# Project Goals Performance

## Technical Requirements

| Customer Need                                         | Technical Need                                                                                                                     | Technical Requirement | Target Value | Passed? |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|---------|
| Precise area navigation                               | Do not come within certain distance of all hazards                                                                                 | $\geq 2\text{cm}$     | 4cm          | ✓       |
| Timely cargo transport                                | Total mission time (maze traversal and drop)                                                                                       | $\leq 18$ minutes     | 12 minutes   | ✗       |
| Detect and characterize hazardous location in a table | Detect a certain percentage of all hazards and display in table                                                                    | $\geq 90\%$           | 100%         | ✓       |
| Generate a map of junctions in area                   | Generate a map of a certain percentage of accuracy displaying junctions                                                            | $\geq 90\%$           | 100%         | ✓       |
| Product is culturally considerate                     | Design uses language independent communication in symbols and human-interaction components in a certain percentage of its displays | $\geq 95\%$           | 100%         | ✓       |
| Cargo delivered status indication                     | Indicate the cargo delivered status in non-hostile manner so that a person from a certain distance away can notice indication      | $\geq 15\text{in}$    | 20in         | ✓       |

# Questions?

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